

Problem 101. Consider the sequence of rational numbers defined by

$$x_1 = \frac{4}{3} \quad \text{and} \quad x_{n+1} = \frac{x_n^2}{x_n^2 - x_n + 1}, \quad \text{for } n \geq 1.$$

If the rational number $\sum_{k=1}^n x_k$ is expressed in lowest terms for $n \geq 1$, show that its numerator is a perfect square.

Solution. First of all, observe that

$$x_1 - 1 = \frac{1}{3} \quad \text{and} \quad x_{n+1} - 1 = \frac{x_n - 1}{(x_n - 1)^2 + x_n},$$

for $n \geq 1$, and so, by induction, all the x_n 's are rational numbers greater than 1.

Next rewrite the recurrence as

$$x_n = \frac{1}{x_{n+1} - 1} - \frac{1}{x_n - 1},$$

for $n \geq 1$, and so

$$\sum_{k=1}^n x_k = \frac{1}{x_{n+1} - 1} - \frac{1}{x_1 - 1} = \frac{x_n^2 - x_n + 1}{x_n - 1} - 3 = \frac{(x_n - 2)^2}{x_n - 1}.$$

Finally, let $x_n - 1 = \frac{a}{b}$, where a, b are positive integers with $\gcd(a, b) = 1$. Then

$$\sum_{k=1}^n x_k = \frac{(a - b)^2}{ab}.$$

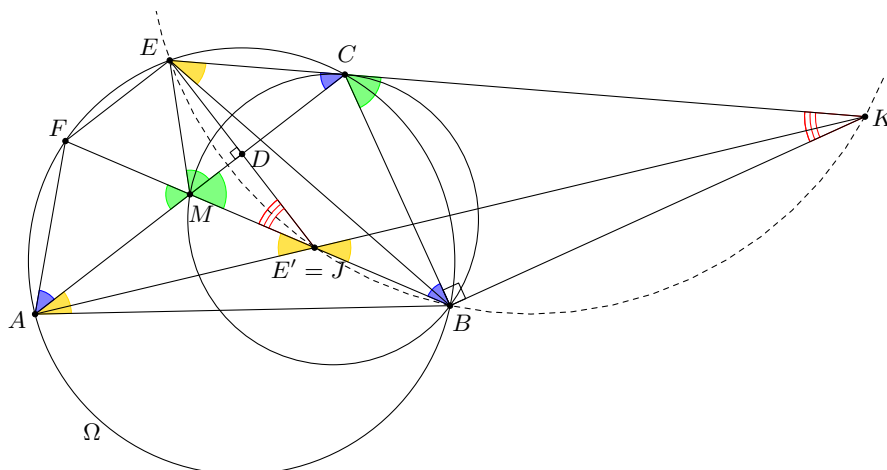
Now, since $\gcd(a, b) = 1$, we have $\gcd(a - b, b) = 1$ and so $\gcd((a - b)^2, b) = 1$. Similarly, $\gcd((a - b)^2, a) = 1$.

Therefore $\gcd((a - b)^2, ab) = 1$ and the result follows. $\square$

Problem 102. Let BM be a median in an acute-angled triangle ABC . A point K is chosen on the line through C tangent to the circumcircle of $\triangle BMC$ so that $\angle KBC = 90^\circ$. The segments AK and BM meet at J .

Prove that the circumcentre of $\triangle BJK$ lies on the line AC .

Solution 1.



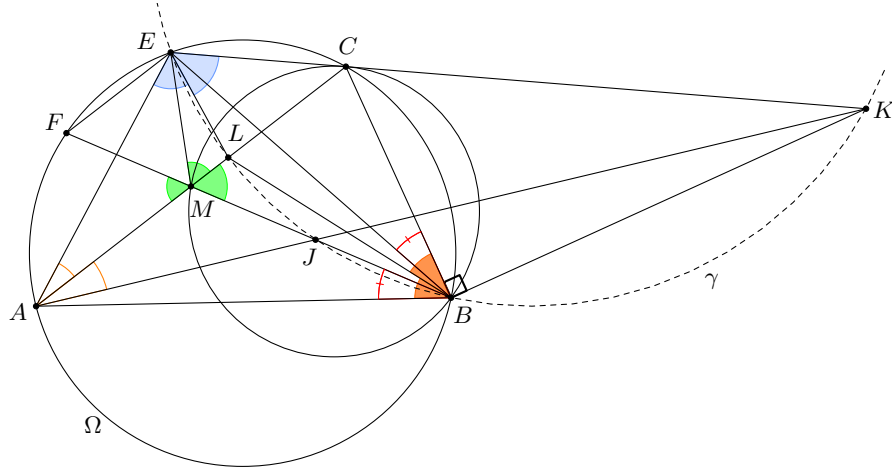
Let Ω be the circumcircle of $\triangle ABC$, and let BM and CK meet Ω again at F and E , respectively. Since CE is a tangent to the circle (BMC) , we have $\angle ACE = \angle MBC = \angle FBC$, so $\widehat{AE} = \widehat{FC}$ and $EF \parallel AC$. By the symmetry in the perpendicular bisector of AC , we get $\angle EMC = \angle FMA = \angle BMC$ and $ME = MF$.

Let E' be the reflection of E in AC ; by the above, E' lies on BM . Since $EE' \perp AC$, we have $\angle EE'M = 90^\circ - \angle BMC = 90^\circ - \angle BCK = \angle BKC$, so the quadrilateral $EE'BK$ is cyclic. Now, since $ME' = ME = MF$, we have $ME' \cdot MB = MF \cdot MB = MA \cdot MC = MA^2$, hence $\triangle MAB \sim \triangle ME'A$, and

$$\angle ME'A = \angle MAB = \angle CAB = \angle CEB = \angle KEB = \angle KE'B.$$

Therefore, E' lies on AK . So $E' = J$, and the circumcentre of $\triangle BJK$ lies on the perpendicular bisector of EE' , i.e. on AC . $\square$

Solution 2.



We present a direct proof of the fact that the points E and J are symmetric to each other with respect to AC . As in the solution above, we introduce the points E and F , and prove that $EF \parallel AC$ and $\angle EMC = \angle FMA = \angle BMC$. This yields that $\angle ABM = \angle EBC$; hence BE is a symmedian in $\triangle ABC$, and quadrilateral $ABCE$ is harmonic, i.e. $\frac{AB}{BC} = \frac{AE}{EC}$.

This means that the interior angle bisectors of $\angle ABC$ and $\angle AEC$ meet at a point L on AC , and that the circle $\gamma = (BLE)$ is an Apollonius circle for the segment AC . On the other hand,

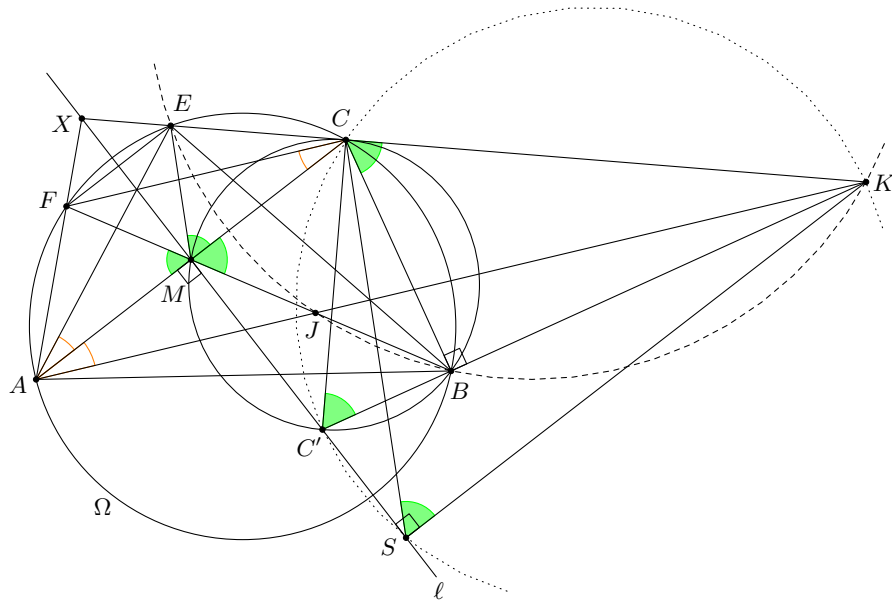
$$\begin{aligned} \angle BLE &= 360^\circ - \angle BCE - (\angle CBL + \angle CEL) \\ &= 360^\circ - \angle BCE - \frac{1}{2}(\angle ABC + \angle AEC) \\ &= 360^\circ - \angle BCE - 90^\circ \\ &= 90^\circ + (180^\circ - \angle BCE) \\ &= 90^\circ + \angle BAE \\ &= 90^\circ + \angle BCK \\ &= 180^\circ - \angle BKE, \end{aligned}$$

hence K also lies on γ . So

$$\frac{AK}{KC} = \frac{AE}{EC}, \quad \text{or equivalently} \quad \frac{AK}{AE} = \frac{KC}{EC}.$$

This yields that AC is an interior angle bisector in $\triangle KAE$. Hence, the points $J = MB \cap AK$ and $E = ME \cap AE$ are symmetric to each other in AC . Since γ is symmetric with respect to AC , the point J lies on γ , which yields what we were to show. $\square$

Solution 3.



We present a yet different direct proof of the fact that the points E and J are symmetric to each other with respect to AC . Again, we introduce the points E and F , and prove that $EF \parallel AC$ and

$$\angle EMC = \angle FMA = \angle BMC.$$

This yields that the point $X = AF \cap CE$ lies on the perpendicular bisector ℓ of AC .

Let C' be the point opposite to C in the circle (BCM) . Then $\angle C'BC = \angle C'MC = 90^\circ$, hence C' lies on BK , as well as on ℓ . Since CK is tangent to (BCM) , we have $\angle C'CK = 90^\circ$. Let S be the projection of K onto ℓ ; due to the right angles at C and S , the quadrilateral $KCC'S$ is cyclic. This yields that

$$\angle KSC = \angle KC'C = \angle BC'C = \angle BMC = \angle CME;$$

so $SC \parallel ME$. Hence, the corresponding sides of the triangles KSC and CME are parallel, and these triangles are homothetic at X . This yields

$$\frac{XF}{XC} = \frac{XE}{XC} = \frac{XC}{XK} = \frac{XA}{XK};$$

so $AK \parallel FC$ and hence

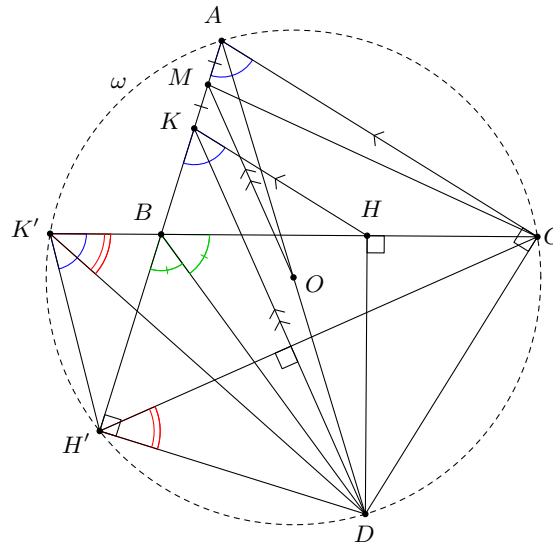
$$\angle KAC = \angle FCA = \angle CAE.$$

So the points $J = MB \cap AK$ and $E = ME \cap AE$ are symmetric to each other in AC . Now one may show, as in Solution 1, that the quadrilateral $EJBK$ is cyclic, and that its circumcentre lies on AC , as desired. $\square$

Problem 103. Let ABC be an acute-angled triangle. The line through C perpendicular to AC meets the external angle bisector of $\angle ABC$ at D . Let H be the foot of the perpendicular from D onto BC . The point K is chosen on AB so that $KH \parallel AC$. Let M be the midpoint of AK .

Prove that $MC = MB + BH$.

Solution.



Reflect K and H across the external bisector BD , obtaining K' and H' , respectively. Then we need to prove that $MC = MB + BH' = MH'$. The reflection also implies that

$$\angle DH'B = \angle DH'A = 90^\circ = 180^\circ - \angle ACD;$$

and

$$\angle CK'H' = \angle BK'H' = \angle BKH = \angle H'AC;$$

so the points A, C, D, H' , and K' lie on a circle ω with diameter AD .

Since $HD \perp K'H$, a simple angle chasing shows that

$$\angle CH'D = \angle CK'D = 90^\circ - \angle K'DH.$$

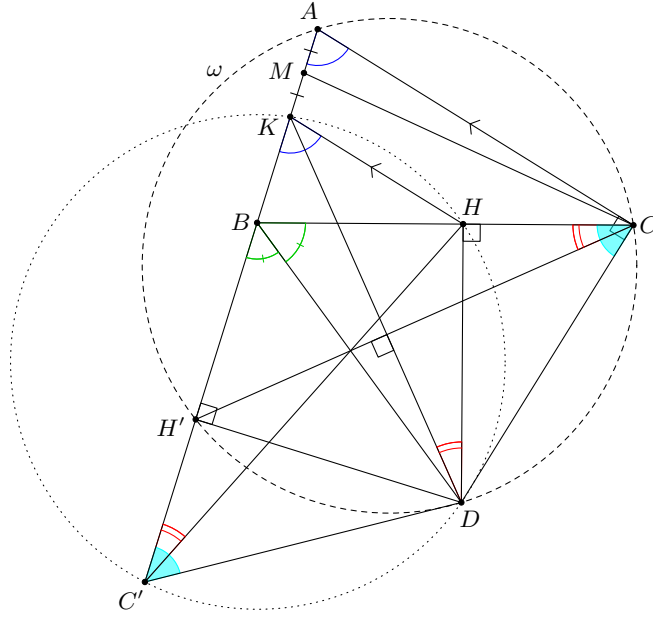
Because of the symmetry, $\angle KDH' = \angle K'DH$, so

$$\angle KDH' + \angle CH'D = 90^\circ,$$

and we conclude that $CH' \perp DK$.

Let O be the center of ω , which is also the midpoint of diameter AD . Then OM is a midline of $\triangle ADK$, and $OM \parallel KD$. Hence the chord CH' of ω is perpendicular to OM , which proves that OM is the perpendicular bisector of CH' . Therefore $MC = MH'$ and we are done. $\square$

Remarks.



(1) There are several variations of the solution above. For example, one may reflect C and H in BD obtaining C' and H' . Since

$$\angle BC'D = \angle BCD = 90^\circ - \angle ACB = 180^\circ - \angle KHD,$$

the quadrilateral $C'DHK$ is cyclic, so

$$\angle KDH = \angle KC'H = \angle BCH'.$$

Since $DH \perp BC$, this yields $KD \perp H'C$. Now one may proceed as above, with the aid of the circle $(H'DCA)$ with diameter AD .

(2) The problem can be solved with the aid of trigonometry, without constructing extra points. In fact, if

$$BC = a, CA = b, AB = c, \quad AM = MK = x, BK = y, \text{ and } BH = z,$$

we need to prove that $CM = x + y + z$, so by the Cosine Formula in $\triangle ACM$ it suffices to show that

$$(x + y + z)^2 = AC^2 + AM^2 - 2 \cdot AC \cdot AM \cos A \iff (y + z)(c + z) = b(y \cos A + a \cos C)$$

which can be verified by computing

$$z = \frac{a \sin \frac{B}{2} \cos C}{\sin\left(\frac{B}{2} + C\right)} \quad \text{and} \quad y = \frac{c \sin \frac{B}{2} \cos C}{\sin\left(\frac{B}{2} + C\right)}.$$

Problem 104. The plane is divided into unit squares by means of two sets of parallel lines. The unit squares are coloured in 1201 colours so that no rectangle of perimeter 100 contains two squares of the same colour.

Show that no rectangle of size 1×1201 contains two squares of the same colour.

Solution. Let the centres of the unit squares be the integer points in the plane, and denote each unit square by the coordinates of its centre. For two unit squares with centres (a, b) and (c, d) , we call $|a - c| + |b - d|$ the *distance* between these unit squares. Note that if the distance between two unit squares is at most 48, then they lie in a certain rectangle of perimeter 100.

Consider the set D of all unit squares (x, y) such that $|x| + |y| \leq 24$. Any translate of D is called a *diamond*.

For any two unit squares that belong to the same diamond, their distance is at most $24 + 24 = 48$. So they belong to some rectangle of perimeter 100. This shows a diamond cannot contain two unit squares of the same colour. Since a diamond contains exactly $24^2 + 25^2 = 1201$ unit squares, a diamond must contain every colour exactly once.

Choose one colour, say, green, and let $a_1, a_2, \dots$ be all green unit squares. Let P_i be the diamond of center a_i . We will show that no unit square is covered by two P_i 's and that every unit square is covered by some P_i .

Indeed, suppose first that P_i and P_j contain the same unit square b . Then their centres are at a distance of at most $24 + 24 = 48$, and hence they lie within the same rectangle of perimeter 100, a contradiction.

Let, on the other hand, b be an arbitrary unit square. The diamond of center b must contain some green unit square a_i . The diamond P_i of centre a_i will then contain b .

Therefore, $P_1, P_2, \dots$ form a covering of the plane in exactly one layer. It is easy to see, though, that, up to translation and reflection, there exists a unique such covering. (Indeed, consider two neighbouring diamonds. Unless they fit neatly, uncoverable spaces of two unit squares are created near the corners: see Figure 1 below.)

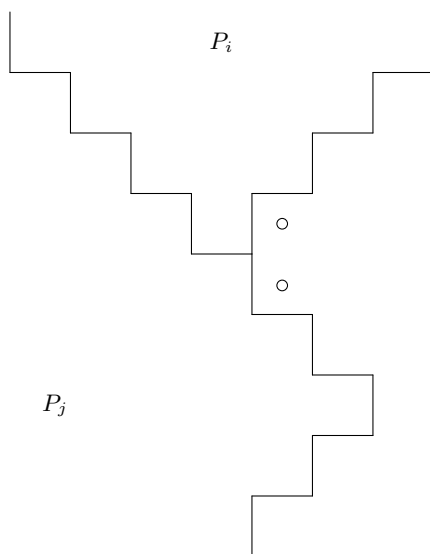


Figure 1

Without loss of generality, then, this covering is given by the diamonds of centers

$$\begin{aligned} & \dots, \\ & \dots, (0, 0), (25, -24), (50, -48), \dots \\ & \dots, (-24, -25), (1, -49), (26, -73), \dots \\ & \dots \end{aligned}$$

It is not hard to see that the centres are those (x, y) such that $24x + 25y$ is divisible by 1201. (See Figure 2 for an analogous covering with smaller diamonds.) It follows from this that no rectangle of size 1×1201 can contain two green unit squares (for example, if 1201 divides both $24x + 25y_1$ and $24x + 25y_2$, then 1201 divides $y_1 - y_2$, and hence $|y_1 - y_2|$ is at least 1201 if $y_1 \neq y_2$), and analogous reasoning works for the remaining colours.

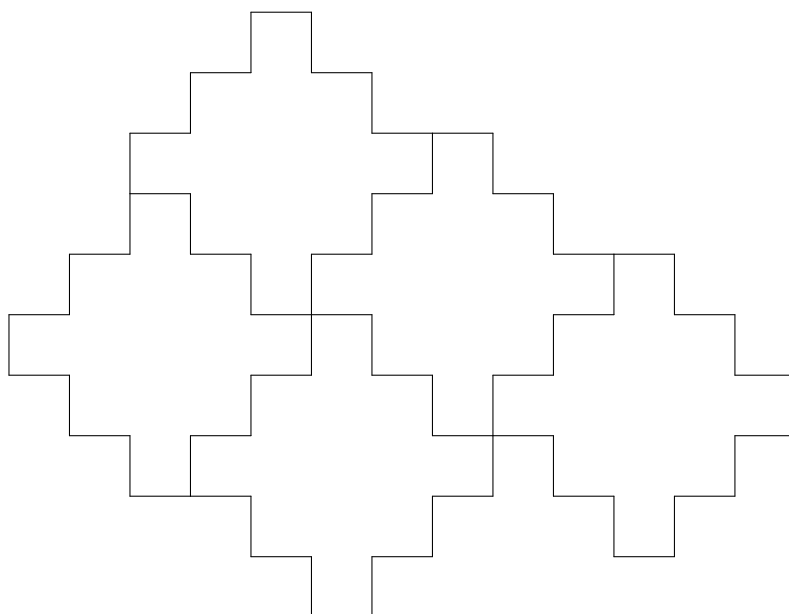


Figure 2

□